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SUPERSEDING
MIL-S-44041(GL)
30 March 1981

MILITARY SPECIFICATION

SHIRT, MAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN

415, DURABLE PRESS

This specification is approved for use by Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for polyester/cotton broadcloth durable press shirt.

1.2 Classification. The shirt shall be of one type in the neck sizes of 13, 13-1/2, 14, 14-1/2, 15, 15-1/2, 16, 16-1/2, 17, 17-1/2 and 18 (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8405

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MILITARY

* MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon-Covered
MIL-C-43594 - Cloth, Interlining, Polyester
MIL-T-43624 - Thread, Polyester, Spun
MIL-C-43992 - Cloth, Broadcloth, Polyester/Cotton, Durable Press
MIL-P-55010 - Plastic Sheet, Polyethylene Terephthalate

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment
MIL-STD-1492 - Provisions for Evaluating Quality of Men's Shirts

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type number 304 - Stainless and Heat Resistant Steel
Type number 316 - Stainless and Heat Resistant Steel

(Application for copies of Steel Product Manual should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Technical Manual of the American Association of Textile Chemists and Colorists

Method 88B (Two photographs) - Photographic Comparative Rating of Single and Double Needle Seams

Method 124-78 - Appearance of Durable Press Fabrics after Repeated Home Launderings

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-204 - Sewing Threads, Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

TEXTILE COLOR CARD ASSOCIATION

Standard Color Card of America

(Application for copies should be addressed to the Textile Color Card Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials (see 6.5).

3.3.1 Basic material. The basic shirt material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992.

* 3.3.1.1 Button stays. The button stay material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992 or as an alternate, a commercial button stay material may be used.

3.3.2 Interlining.

3.3.2.1 Collarstand, shoulder loops and pocket flaps. The material for interlining the collarstand, shoulder loops and pocket flaps shall be polyester fiber cloth conforming to MIL-C-43594.

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3.3.2.2 Collar. The material for interlining the collar shall be an undyed, preshrunk, 100 percent cotton, plain weave cloth which has a uniformly distributed dot system of high density polyethylene fusible adhesive applied to one side in accordance with commercial practice. The fusible interlining shall meet the following requirements when tested as specified in 4.4.1.

Weight, oz. per sq. yd.	4.5 ± 0.5
Weave Plain	
Threads per inch (min) Warp	44
Filling	40
Shrinkage, percent (max) Warp	2.0
Filling	2.0

In addition, the fusible interlining shall meet the following requirements after it has been fused to the basic cloth, when tested as specified in 4.4.1.

Stiffness, inch lbs Warp	0.010 - 0.020
Filling	0.003 - 0.008
Adhesion after 10 launderings oz, per inch (min)	20

3.3.2.2.1 Fusing conditions. The fusible interlining shall be fused to the basic material on a fusing press (see 6.8). The adhesion of the unlaundered sample shall be a minimum of 30 ounces per inch when tested as specified in 4.4.2.1.

3.3.3 Labels. Each shirt shall have an identification label, size label and instruction label conforming to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.3.3.1 Identification label. The identification label shall conform to type VI, class 1.

3.3.3.2 Size label. The size label shall conform to type IV, class 2.

* 3.3.3.3 Instruction label. The instruction label shall conform to type VI, class 3, except that the label cloth shall be dyed Lemon Yellow, Cable No. 70205, in conformance with the Standard Color Card of America. The label shall contain the following information.

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SHIRT, MAN'S, SHORT SLEEVE, POLYESTER/COTTON ARMY GREEN 415, DURABLE PRESS

Washing Instructions

1. Wash with moderately hot water (115°F) using soap or mild detergent.

   a. For machine washing. Use wash and wear cycle.
   b. For hand washing. Do not wring or twist.

2. DO NOT STARCH OR BLEACH

3. Dry in tumble dryer at warm setting. Remove immediately at end of drying. Shape and hang.

4. To drip dry, remove from water before last spin cycle. Shape and hang.

5. If touch up is required, iron at a moderate or synthetic material heat setting.

DO NOT REMOVE THIS LABEL

3.3.3.4 Combination identification-instruction label. The identification label and instruction label may be combined into a type VI, class 15 label, except the 1/2 inch space between the labels may be reduced to 1/4 inch. When combined, the color of the label shall conform to the requirements of the instruction label (3.3.3.3) and the item nomenclature shall not be repeated. The coating requirements of the type VI, class 3 label shall apply.

* 3.3.4 Thread. The thread for seaming, stitching, buttonholes and button sewing shall be cotton-covered polyester, ticket Nos. 70, 2 or 3 ply and 100, 2 or 3 ply conforming to MIL-T-43548. Spun polyester thread conforming to class 1 or 2 of MIL-T-43624 may be substituted for the cotton-covered polyester thread as shown below. The thread shall be stabilized so as not to shrink in boiling water more than 1.5 percent when tested as specified in table IV.

Spun polyester	Cotton-covered polyester
70, 2 or 3 ply	70, 2 or 3 ply
100, 2 ply	100, 2 or 3 ply

3.3.4.1 Color and colorfastness. The thread shall be White AH, C.A. 66050. The dyed thread shall show fastness to light, perspiration, and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light, perspiration and laundering.

3.3.5 Buttons. The buttons shall conform to type II, class K, style 25, 19 line of V-B-871. The color of the buttons shall be shade White BA, C.A. 62031.

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3.3.6 Collar stays. The collar stays shall be made from polyethylene terephthalate plastic sheet conforming to MIL-P-55010, except the thickness shall be 0.014 ± 0.001 inch, the color shall be natural (clear or opaque), and the transparency and optical properties shall not apply. Either both ends shall be rounded or pointed with the points rounded, or one end rounded and one end cut at any angle with the point rounded. Testing shall be as specified in 4.4.1.1.

3.3.6.1 Collar stay configurations.

3.3.6.1.1 Unslotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 3-3/8 to 3-1/2 inches long.

3.3.6.1.2 Slotted stays. The stays shall measure 1/2 ± 1/32 inch wide and 3-3/8 to 3-1/2 inches long. The stay shall have two slots 1-3/8 ± 1/32 inches long, centered along the length of the stay. Each slot shall be 1/32 inch wide and spaced approximately equal in distance across the width of the stay.

3.4 Design. The shirt is a short sleeve dress style with a convertible collar. The shirt has a seven button front, two breast pockets, with button through flaps, pen pocket inside the left breast pocket, and shoulder loops. The collar has a fusible interlining; the collarstand, shoulder loops and pocket flaps have polyester cloth interlinings; collars have stays; and the shirt is of a safety-stitch construction and has a yoke (see figure 1).

3.5 Patterns. Standard patterns will be furnished by the Government and, unless otherwise indicated, provide an allowance of 1/4 inch for all seams except 3/8 inch for setting the pen and breast pockets. The standard patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns.

3.5.1 List of pattern parts. The components of the shirt shall be cut from the material as specified below, in accordance with the pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Basic material	Front	2
	Back	1
	Yoke	2
	Sleeve	2
	Shoulder loop	4
	Collar	2
	Pocket 2/	2
	Pocket flap 2/	4
	Pen pocket	1
	Stay holder 1/	2

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TABLE I. List of pattern parts (cont'd)

Material	Nomenclature of pattern parts	Cut parts
Polyester fiber interlining cloth	Shoulder loop interlining	2
	Pocket flap interlining	2
	Top collarstand interlining	1
Fusible interlining cloth	Collar interlining	1

1/ If used.

2/ 1/8 inch may be added to three sides (see operation 1.e).

* 3.6 End item seam appearance, bond strength of fusibles and adhesion of ultrasonically fused collar stays.

3.6.1 Seam appearance rating. After the finished shirt has been laundered five times (see 4.4.4), the appearance rating of the seams and stitchings shall be as follows:

Part	Minimum average of 3 observations	Minimum individual observations
Fused collar	3.8	3.0
Shoulder loops and yoke	3.6	3.0
Armholes, cuffs, pockets, sleeve hems, and pocket flaps	3.0	2.7
Side and underarm seam	3.2	2.0

3.6.2 Bond strength of fusibles. After the shirt has been laundered 10 times, it shall show no bubbling or delamination of the fused plies of material when tested as specified in 4.4.4. There shall be no showing of the "strike-through" of the adhesive after fusing.

* 3.6.3 Adhesion of ultrasonically fused collar stays. After the shirt has been laundered 10 times, the shirt shall show no delamination of the ultrasonically fused collar stays when tested as specified in 4.4.4. There shall be no delamination of any fused points.

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* 3.7 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from the written document appear in figure 1, the written document shall govern.

3.7.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the shirt.

3.7.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or over stitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.7.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/4 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/4 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/4 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.7.1.2 Type 401, 502, 503, 515, 516 and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.7.1.1.1 a. and b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the material sewn. 301 stitching may be used to repair 401 stitching.

* 3.7.1.3 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches (minimum). Bartacks shall be free from thread breaks and loose stitching.

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3.7.2 Buttonholes. The buttonholes shall be straight-cut, purl or whip type and the finished cut length shall be 1/2 to 9/16 inch long. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. When a purl-type buttonhole is used, the purling shall be on the outside of the shirt, except for the neck closure buttonhole which shall have the purling on the inside of the shirt. All loose thread ends shall be trimmed or fused to within 1/8 inch of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

* 3.7.3 Automatic stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

* 3.7.3.1 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

* 3.7.4 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirts.

3.8 Manufacturing operations requirements. The shirt shall be made in accordance with operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished shirt.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut shirts in strict accordance with patterns furnished which show size, directional lines, marks, and notches for proper assembly. Directional lines shall be placed in the warp direction. The fronts shall be cut on selvage. When fronts of shirts are cut from the uneven edge of the lay, they shall be blocked out and relaid so that the width of the facings shall be exactly as the pattern. When selvageless basic cloth is utilized (i.e., cloth woven or shuttleless looms), the facings shall be overedge stitched as indicated in operation 9., and blocking will not be required. The use of drill holes are prohibited except for the pen pockets and sewn labels, providing the drill holes are covered by the outer fabrics or label respectively, otherwise exposed drill holes shall be considered to be holes in the fabric. b. Cut component parts from same piece of the basic material, except the undercollar, stay holders, button stay, pen pocket, the under parts of the pocket flaps, shoulder loops and yoke which may be cut from ends. Parts cut from ends shall approximate the shade of the other parts. c. Cut the fusible and polyester fiber interlinings in accordance with directional lines on patterns. d. Cut button stay from basic material 3/4 + 1/8 inch diameter or square.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) e. The pocket and flap may be cut 1/8 inch larger on three sides (excluding top of pocket and top of flap); however, the finished dimensions shall be maintained.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1492 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Component marking. a. Mark, and ticket all components of the shirt except those parts cut from ends to insure correct size, proper assembly, and uniform shade throughout the shirt. b. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the shirt and the numbers are covered by the seam allowance. c. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew-on type tickets. 3. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labeling.						
	a. Center size marking on top collarstand (1/4 inch off center tolerance), so that on the finished shirt, the bottom of the marking will not be less than 1/4 inch or more than 1/2 inch from the seam joining the collarstand to the neck.						
	b. Position the yellow instruction label on the outside right front (as worn) so that on the finished shirt the bottom of the label shall be 1 to 2 inches from the hemmed bottom of the shirt and the side of the label shall be approximately 2-1/4 to 3 inches from the front folded edge and stitch on all four sides. The stitching shall not be through the printing.	301	LSbj-1	8-14	70-2	70-2	
	c. Position and stitch the identification label as in "b" above except it shall be placed on the left front.	301	LSbj-1	8-14	70-2	70-2	
	d. The yellow combination label, when applicable, shall be positioned and stitched as in "b" above.	301	LSbj-1	8-14	70-2	70-2	
5.	Make collar.						
	a. Unslotted collar stay holder construction - (when used).						
	(1) Position fusible interlining to top collar 1/16 inch from top and side edges and fuse (see 3.3.2.2.1).						

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					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. (cont'd)						
	(2) Position stay holder to under collar with side and bottom edges even and stitch along bottom edges to hold stay holder in place.	301	SSv-2	10-14	70-2	70-2	
	(3) Form a tunnel for the collar stays by stitching the stay holder to under collar with two rows of stitching 7/16 to 1/2 inch apart according to marks on pattern. The collar points shall be aligned with the center of the two rows of stitching and the stitching shall extend to the raw edges.	301	SSv-2	10-14	70-2	70-2	
	(4) Insert collar stays into the tunnels so tapered ends will abut the top edge stitching when stitched as specified in operation 5.g. Stitch or tack tunnel at bottom of stay holder.	301 or 101	SSv-1 SSv-1	10-14 6-8	70-2 70-2	70-2	
*	b. Unslotted collar stay ultrasonically fused construction - (when used).						
	(1) Position fusible interlining to top collar 1/16 inch from top and side edges and fuse (see 3.3.2.2.1).						
	(2) Position collar stays according to marks on pattern so tapered ends will abut the top edge stitching when stitched as specified in operation 5.f.						
	(3) Ultrasonically fuse the collar stays to the under collar with two rows of fusing points the full length of the stay. Each row shall contain nine to eleven fusing points per inch.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make shoulder loops.						
	Finished appearance. The shoulder loops shall finish smooth and flat and uniform in construction and placement. The buttonholes shall be clean cut and correctly positioned. The shoulder loops shall measure finished 2 +0, -1/8 inches wide at the armhole seam, tapering to a width of 1-1/2 +0, -1/8 inches at pointed end.						
	a. Position the shoulder loops face to face with interlining on top and stitch the three plies of the shoulder loop and interlining around all edges except armhole end.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	70-2 70-2	70-2 100-2	
	b. Turn loop, work out points and edges, and edge stitch along sides and point, 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	70-2	70-2	
	c. Make a horizontal buttonhole in the center of each loop 1/8 inch off-center tolerance, with the inside cut edge of the buttonhole 3/8 to 1/2 inch from point of loop (see 3.7.2).	304	Whip or purl types	42-52 per buttonhole including tack	70-2	70-2	
	d. Press shoulder loops with a hot head pressing machine (see operation 22).						
7.	Make pocket flaps.						
	a. Position the pocket flaps face to face, align the interlining on top of basic with sides and bottom even, and seam the flaps along bottom and sides; turn, work out points and edge stitch 3/16 to 1/4 inch from side and bottom edges.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b)	10-14 10-14 10-14	70-2 70-2 70-2	70-2 100-2 70-2	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make pocket flaps. (cont'd)						
	b. Make one vertical buttonhole in the center of each flap (1/8 inch off-center tolerance) with the lower inside cut edge 3/8 to 1/2 inch from bottom edge of flap (see 3.7.2).	304	Whip or purl types	44-52 per buttonhole including tack	70-2	70-2	
	c. Press pocket flaps with a hot head pressing machine (see operation 22).						
8.	Hem breast pocket.						
*	Fold under top edge of pocket as indicated by pattern marks with the raw edge turned under 1/4 inch and stitch 1/16 inch from the folded edge. The hem shall finish 1-1/8 ± 1/8 inches wide.	301	EFb-1	10-14	70-2	70-2	
		or 401	EFb-1	10-14	70-2	100-2	
	or						
	Overedge stitch top raw edge of pocket and turn top of pocket down to form hem. The hem shall finish 1-3/8 ± 1/8 inches wide.	502 or 503	EFd-1	10-14	100-2	100-2	
			EFd-1	10-14	100-2	100-2	
9.	Make fronts.						
	If selvage is cut, frayed or selvageless cloth is utilized (i.e. cloth woven on shuttleless looms) overedge back edge of facings.	502 or 503	EFd-1	10-14	100-2	100-2	
			EFd-1	10-14	100-2	100-2	
10.	Make and set pen pocket.						
*	a. Fold under top edge of pen pocket as indicated by pattern marks with the raw edge turned under 1/4 inch and stitch 1/16 inch from the folded edge. The hem shall finish 1 + 1/8 inches wide.	301	EFb-1	10-14	70-2	70-2	
		or 401	EFb-1	10-14	70-2	100-2	

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					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make and set pen pocket. (cont'd)						
	or						
	Overedge stitch top raw edge of pocket and turn top of pocket down to form hem. The hem shall finish 1-1/4 ± 1/8 inches wide.	502 or 503	EFd-1 EFd-1	10-14 10-14	100-2 100-2	100-2 100-2	
	b. Turn in the side edges of the pen pocket 3/8 inch, and stitch to the left front of the shirt as indicated on the pattern. The finished pen pocket shall be 2-1/4 ± 1/8 inches wide. (NOTE: The bottom of pocket shall not be stitched.)	301	LSd-1	10-14	70-2	70-2	
	c. Form two compartments with a row of vertical stitching at the center ± 1/8 inch through the shirt front from top to bottom of pocket.	301	SSv-1	10-14	70-2	70-2	
11.	Attach pockets and flaps.						
	Finished appearance. The pocket shall be parallel to the front edge of shirt. The pocket depth at center for all sizes shall finish 5-7/8 ± 3/16 inches and the finished widths shall be: 4-1/2 ± 3/16 inches wide for sizes 13 through 14-1/2. 5 ± 3/16 inches wide for sizes 15 through 16-1/2. 5-1/2 ± 3/16 inches wide for sizes 17 through 18.						
*	a. Stitch pockets on fronts, as indicated by marks on patterns, 1/16 to 1/8 inch from edge, continuing stitching 1/4 to 3/8 inch across top of pockets and 1/2 to 5/8 inch diagonally down to the line of 1/16 to 1/8 inch stitching. Do not catch front facing in stitching. The pockets may be precreased.	301 301	LSd-1 and SSv-1	10-14 10-14	100-2 100-2	70-2 70-2	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
11.	Attach pockets and flaps. (cont'd)						
	b. Press right and left fronts with a hot head pressing machine (see operation 22).						
	c. Position the raw edge of flap on line with marks on pattern and stitch, 1/8 to 3/16 inch from raw edge. Turn down flap and raise stitch 3/16 to 1/4 inch from turned edge with the ends of the flap tacked. Pockets and flaps shall be in parallel alignment with each other and front edge of shirt and ends of pocket openings shall be covered by flap. The distance between the top of pocket and the top folded flap edge on the finished shirt shall be 3/4 ± 1/8 inch.	301	LSb1-2	10-14	70-2	70-2	
12.	Attach yoke.						
	Finished appearance. The finished yoke shall be without twists or puckers. There shall be no fullness between the plies of material.						
	a. Position back of shirt between the two plies of yoke and with the back and yoke edges even, stitch all plies together.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 100-2	
	b. Press back and yoke, with yoke plies and back in the finished full length position, smooth, flat and even along the stitching, with heated iron at a permanent press (moderate) setting. (See operation 22.a.)						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Join shoulder seam. Position fronts between the two plies of yoke and with front and yoke edges even, stitch all plies together. The stitching shall not show on the finished shirt.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 100-2	
14.	Set shoulder loops. Position shoulder loop so point of loop is 1/4 ± 1/8 inch back of shoulder seam, 3/8 ± 1/8 inch from collar joining finished seam with front edge of loop at armhole opening 1 ± 1/8 inches forward of shoulder seam. Stitch 1/8 to 3/16 inch from raw edge of armhole opening and trim excess even with opening. The excess of the shoulder loop may be trimmed in operation 15.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 100-2	
15.	Set sleeves. Set sleeve to armhole and join with safety stitch catching the raw edges of the shoulder loop in the seaming.	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 100-2	100-2 100-2 (overedge)	
16.	Join side and underarm seams. Safety stitch side and sleeve seams in a continuous operation. Armhole seam shall meet at underarm seam, 1/4 inch tolerance	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 100-2 (over-edge)	100-2 100-2	

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TABLE II. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Join collar to shirt. Finished appearance: The collar shall be carefully joined to the shirt, with notches matching, without distorting the top or under collars. Collar ends shall not be uneven in length by more than 1/8 inch. In stitching down top collar, the stitching shall be not more than 1/8 inch on or off the opposite under collar.						
	a. Fold fronts at notches to form facings. Position the collar assembly between the front and facing at the collar notch and stitch from folded edge to within 3/4 ± 1/8 inch of inner edge of facing. Notch collar thru front and facing, continue stitching under collar to neck of shirt.	301	LScf-2(a)	10-14	70-2	70-2	
	b. Turn facings and work out steps. Turn under unsewn portion of top collar and stitch to neck line 1/16 to 1/8 inch from turned edge, catching the ends of the facings in the stitching.	301	LScf-2(b)	10-14	70-2	70-2	
18.	Hem shirt bottom. Hem bottom of shirt with the hem extending across fold of fronts. Hem shall finish 1/8 to 5/16 inch wide. The side seams shall be turned in the same direction as sleeve hems.	301	EFb-1	10-14	70-2	70-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Hem sleeve.						
*	Turn under bottom edge of sleeve, with raw edge turned in, and stitch 1/16 to 1/8 inch from edge. The finished hem shall measure 1 ± 1/8 inch wide. (NOTE: The hemming of sleeve may be performed before or after joining underarm seam.)	301	EFb-1	10-14	70-2	70-2	
		or 401	EFb-1	10-14	70-2	100-2	
	When sleeve is hemmed prior to joining underarm seams, place a horizontal bartack across seam allowances, at the end of the underarm seam 3/16 to 1/4 inch from bottom of sleeve.	Bartack	--	28 per bartack	70-2	70-2	
20.	Make buttonholes.						
	a. Make six vertical buttonholes in the left front of shirt through the facing as indicated by marks on pattern positioned 3/4 ± 1/16 inch in from front edge (see 3.7.2).	304	Whip or purl types	42-52 per button-hole in-cluding tack	70-2	70-2	
	b. Make one horizontal buttonhole at top on left front of shirt through the facing with front inside cut edge of buttonhole 1/2 ± 1/16 inch from front edge and 3/8 ± 1/16 inch below top of facing (see 3.7.2). The buttonhole shall be parallel to the facing top.	304	Whip or purl types	42-52 per button-hole in-cluding tack	70-2	70-2	
21.	Clean shirts.						
	a. Before folding and packaging, shirts must have all thread ends trimmed and removed, be free of residual fiber particles, lint and dust, or other foreign substances, and spots and stains removed.						
	b. Remove all shade tickets.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Pressing.						
	a. As an option to operations 11.b., and 12.b., a hot head pressing machine capable of pressing the entire shirt after fabrication may be used (see 6.7).						
	b. Press the bottom of shirt, side seams, sleeve hems and underarm seams on a hot head pressing machine.						
*	NOTE: The use of steam inflated bag type pressing machine is prohibited. The hot head pressing machine cited in operations 6.d., 7.c., and 11.b., shall be equipped with controls for accurate temperature (350 to 370°F), time (minimum 10 seconds) and pressure (100 psi gauge).						
23.	Sew on buttons.						
	Finished appearance. The buttons shall be aligned and securely sewn in the center of the buttonholes ± 1/16 inch on front of shirt, shoulder loops and flaps. Collar ends shall be centered equidistant from center of shirt front and shall open not less than 1/2 or more than 1 inch, when collar is buttoned.						
	a. The top button shall be sewn with a back reinforced button stay consisting of basic or alternate material cut as required in operation 1.d. or as per 3.3.1.1. Position the top button with button stay between right front and facing 5/8 ± 1/16 inch from front edge and sew. The stitching shall not penetrate the facing material.	301 or 101	-- --	14-16 per button 14-16 per button	70-2 70-2	70-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Set on buttons. (cont'd)						
	b. Sew remaining six buttons on the right front 3/4 + 1/16 inch from the front edge through the front facing.	301 or	--	14-16 per button	70-2	70-2	
		101	--	14-16 per button	70-2		
	c. Sew one button on each breast pocket through hem to correspond with flap buttonhole.	301 or	--	14-16 per button	70-2	70-2	
		101	--	14-16 per button	70-2		
	d. Sew one button on each shoulder seam allowance through yoke to correspond with the buttonhole in the shoulder loop. The shoulder loops shall lay smooth and flat without puckers or twists when they are buttoned.	301 or	--	14-16 per button	70-2	70-2	
		101	--	14-16 per button	70-2		
24.	Touch-up pressing and closing shirt.						
	a. When required, touch-up fronts, backs, sleeve hems and collars, before folding for packaging with a heated iron at a permanent press (moderate) setting.						
	b. Button shirt fully.						

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3.9 Finished measurements. The finished measurements shall conform to table III.

TABLE III. Finished measurements (inches)

Size	Neck size 1/	1/2 Chest 2/	Back length 3/	Sleeve length 4/	Collar opening 5/
13	13-1/2	19	31	6-1/2	1/2 to 1 inch (applicable to all sizes)
13-1/2	14	20	31-1/4	6-1/2	
14	14-1/2	21	31-1/2	6-1/2	
14-1/2	15	22	31-3/4	6-1/2	
15	15-1/2	23	32	6-1/2	
15-1/2	16	24	32-1/4	6-1/2	
16	16-1/2	25	32-1/2	6-1/2	
16-1/2	17	26	32-3/4	6-1/2	
17	17-1/2	27	33	6-1/2	
17-1/2	18	28	33-1/4	6-1/2	
18	18-1/2	29	33-1/2	6-1/2	
Tolerance:					
Plus	1/8	3/8	1/2	1/4	
Minus	1/8	3/8	1/4	3/8	

1/ Neck measurements taken from center of button to front cut-end of buttonhole, with collar in flat position.

2/ Chest measurement taken from folded edge to folded edge at base of armhole with shirt buttoned (see A of figure 1).

3/ Back length measurement taken at center back from collar setting seam to bottom edge of shirt (see B of figure 1).

4/ Sleeve length measurement taken from base of armhole along the inseam to bottom of sleeve (see C of figure 1).

5/ Collar opening measurement shall be taken with the shirt buttoned, placed on a flat surface and without distorting collar, measure across collar from ends of collar at joining seam.

3.10 Workmanship. The finished shirts shall conform to the quality of product established by this document. As a final step in the contractor's production control plan before formation of a lot, each shirt shall be examined, buttoned, pressed and laid out flat on a table prior to folding for packaging. A shirt containing a selected defect shall not be included in the end item lot. Selected defects are those defects listed in MIL-STD-1492, indicated by an asterisk (*). The occurence of defects shall not exceed the applicable point value or defect limit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4.3).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in MIL-STD-1492 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition to testing provisions contained in subsidiary documents, testing shall be performed on the components listed in table IV for characteristics noted. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table IV, shall be followed. The sample size shall be as specified in table V. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The unit for expressing lot size, and the sample unit for testing each component shall be as follows:

Component	Lot size expressed as	Sample unit for testing
Fusible interlining for collar leaf	yards	2 yards full width
Fusible interlining fused to basic cloth	yards	1 yard full width
Thread	Holders	1/

1/ One holder or holders containing sufficient thread length to provide thread for all applicable tests.

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TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Fusible interlining for collar leaf	Weave	3.3.2.2	Visual
	Weight	3.3.2.2	5041
	Threads per inch	3.3.2.2	5050
	Shrinkage	3.3.2.2	5556
Fusible interlining fused to basic cloth	Stiffness	3.3.2.2	5202
	Adhesion after 10 launderings	3.3.2.2	5951 1/
Thread	Shrinkage	3.3.4	ASTM D204 2/

1/ Material shall be laundered as specified in 4.5.1.3.1 prior to testing for adhesion.

2/ Shrinkage in dry heat shall not apply.

TABLE V. Sample size

Lot size	Sample size
Up to 22,000 (inclusive)	8
22,001 and over	13

4.4.1.1 Examination of collar stays. Eight collar stays shall be randomly selected from each lot and examined for conformance to configuration and dimensional requirements in 3.3.6. Any nonconforming collar stay found in the sample shall be cause for rejecting the lot represented.

4.4.2 In-process inspection.

* 4.4.2.1 Adhesion testing of fusibles. Prior to fusing each 1,000 parts, a sample of equivalent area to a part shall be fused for visual and adhesion testing. After visual examination for bubbling, delamination and bleed through, the sample shall be cut to a 1-inch width and peeled in the warp direction using a spring scale or suitable device capable of measuring the peel strength in ounces. Any sample failing the minimum peel strength requirment specified in 3.3.2.2.1 or showing any bubbling, delamination, or bleed-through shall be cause for rejection of the previous 1,000 fused parts. The rejected lot shall not be used in assembly of the shirts. No fusing shall be performed after the rejection of a 1,000 part lot until the machine has been properly adjusted to produce a passing sample. The rejected lot shall then be replaced with a shade matched lot which has been fused correctly. Rejection of the previously fused parts shall not apply to the first sample tested after machine warm-up.

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* 4.4.3 Point count inspection. Sampling and inspection provisions for end item examination of dress "D" shirts, dimensional examination, and packaging inspection shall be performed in accordance with MIL-STD-1492 except that for end item examination, the defects for knots or slubs on the outside, and the opening between collar ends, shall be as follows:

Defect	Point value
Knot larger than B, in zone X.	3
Knot larger than C, in zone Y.	1
Slub larger than E, in zones X and Y (all areas except collar).	1
Slub larger than C, in collar.	3
Collar ends open less than 1/2 inch or more than 1 inch when buttoned.	3

* 4.4.4 End item testing. The shirts shall be tested as specified in table VI. All test reports shall contain the individual values utilized in expressing the final results. All requirements shall be applicable to the sample unit. The sample size shall be as specified in table V. The lot size shall be expressed in units of shirts and the sample unit shall be two finished shirts. One shirt shall be used to evaluate the appearance requirements and the other shirt shall be used to evaluate for bubbling and delamination of collar and delamination of collar stays. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method
Seam appearance rating	3.6.1	4.5.1
Bubbling or delamination of collars	3.6.2	4.5.1
*Adhesions of ultrasonically fused collar stays	3.6.3	4.5.1

4.5 Methods of inspection.

* 4.5.1 Appearance of seams, bubbling or delamination of collar, adhesion of fusible interlining and untrasonically fused collar stays.

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4.5.1.1 Apparatus and material used for laundering. The following apparatus and material is required for laundering:

a. Automatic washing machine (see 6.7).
b. Automatic dryer (see 6.7).
c. AATCC Standard Detergent 124 or equivalent.

4.5.1.2 Apparatus and material used for evaluation. The following apparatus and material is required for evaluation of samples for seam appearance:

a. Overhead lighting procedure apparatus as specified in AATCC Test Method 124-78.
b. AATCC photographic standards for evaluating single needle stitching.
c. Evaluation room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black, or blackout curtains may be hung on each side of the viewing area.

4.5.1.3 Procedure.

4.5.1.3.1 Laundering and conditioning. If the shirts or material selected as test samples weigh less than 4 pounds, ballast to make up a standard 4-pound load shall be added to the test samples. The ballast shall consist of durable press shirts that are approximately the same weight as the test shirts or pieces of durable press fabric that are similar to the fabric of the test samples. The pieces shall be hemmed and measure 3 feet by 3 feet. The front of all shirts in the 4-pound load shall be unbuttoned. The load shall be placed in the washing machine. Add 90 grams of detergent to the washing machine. The washing machine shall be set for 10 minutes on the wash and wear (permanent press) phase high water level. The washing machine shall be filled with water with a hardness not greater than 50 parts per million (PPM), and the wash water shall be at a temperature of 115 $\pm$ 5°F with a cold rinse. Immediately after the washing machine has completed the final spin phase the entire load shall be removed, untangled, and placed in the dryer. The dryer shall be operated at a temperature of 140 to 160°F until the entire load is dry, continue tumbling 10 minutes with the heat turned off (cool-down phase). Remove the load immediately after the machine stops. Test sample material for adhesion testing shall be subjected to 10 complete cycles. Test sample shirts for seam appearance rating shall be subjected to 5 complete cycles. Test sample shirts for bubbling and delamination of collars shall be subjected to 10 complete cycles. After completion of the last cycle, the test sample shall be conditioned for a minimum of 2 hours at 70 $\pm$ 10°F and at a relative humidity of 50 to 65 percent. (NOTE: During the interruptions between the cycles and during the conditioning and the evaluations, test shirts shall be placed on wooden or plastic hangers and shall be buttoned at the front).

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4.5.1.4 Evaluation of the appearance of seams. After the test shirts have been laundered and conditioned as specified in 4.5.1.3.1, the shirts shall be evaluated by three trained observers. The trained observers, independent of the other observers, shall evaluate each of the shirts as follows: The shirts shall be placed on a clothes hanger with the buttons fastened and the hanger suspended on the viewing board in such a manner that the shirt lies flat on the surface of the board. The photographic standards for single needle stitching shall be placed on the viewing board. The observer shall stand directly in front but 4 feet from the test shirt. The observer shall then compare the test shirt, in each of the areas specified in 3.6.1, to the photographic standards and shall assign a rating value to each of the specified areas. The rating value assigned to an area by the observer shall be the numerical value of the photographic standard that most nearly matches the area. No estimation or ratings falling between standards are allowed. Each rating value assigned by each of the observers shall be recorded. The average of the rating values assigned by each observer to the same area shall be the rating value for that area.

4.5.1.5 Bubbling and delamination of collars. After the test shirts have been laundered and conditioned as specified in 4.5.1.3.1, the collars shall be examined visually for bubbling and delamination.

* 4.5.1.6 Adhesion of the ultrasonically fused collar stays. After the test shirts have been laundered and conditioned as specified in 4.5.1.3.1 and collars examined for bubbling and delamination, the collar shall be opened and the stays examined visually for delamination.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each fully buttoned shirt shall be neatly folded to measure approximately, 14-1/2 inches by 10-1/2 inches. The folded shirt shall be secured with stainless steel (AISI type 304 or 316, see 2.2) pins, aluminum clips, or plastic fasteners. Each shirt shall be inserted in a snug-fitting flat style clear polyethylene film bag of 0.00125 inch thickness (± 25 percent tolerance). The polyethyelene bag shall be formed with heat sealed seams that are straight and continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. A 1/4 inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape. Alternatively, the final closure of polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Commercial. The shirts shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Seventy-two shirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type 1, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with MIL-STD-1492. Toward the end of the contract, or when there are less than the requirement amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2 Level B packing. Seventy-two shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth within the shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type 1, grade B of UU-P-268. Each container shall be closed in accordance with method II as specified in PPP-B-636, except that inspection shall be in accordance with MIL-STD-1492. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with MIL-STD-1492.

5.2.3 Commercial packing. Shirts, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

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5.3.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The short sleeve shirts are intended for wear by male military personnel of the Department of the Army.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Size required (see 1.2).
(c) When a first article sample is required (see 3.2, 4.3 and 6.4).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample or initial production item. The contracting officer should specify the appropriate type of first article and the number of units to be furnished.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.6 Seam appearance. In order to avoid puckering and to produce the flattest seams possible, consideration must be given to the following:

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate hole should be no larger than necessary to permit free entry of the needle and to prevent flagging.

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c. The finest feed (approximately 18 teeth) possible should be used with a pressure foot to exactly match the shape and size of the feed. The face of the pressure foot (underside next to the feed on the machine) should be highly polished to eliminate roughness and reduce drag. Diamond shaped feed teeth have been used quite successfully.

d. The lightest pressure foot pressure should be used to prevent stretching of the top ply of material.

e. In the safety stitch machine (515, 516 and 519), the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the edge of the fabric.

f. In stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

g. When sewing with stitch type 401 (including the seaming row of the safety stitches), the thread ratio (needle and looper) should be adjusted to produce a stitch with 60 to 70 percent needle thread, and both needle and looper thread tension should be held to the minimum.

h. Thread takeup mechanism should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free of burrs, cuts and rough spots.

6.7 Apparatus. The recommended apparatus for 4.5.1.1 a. and b., and for pressing in operation 22 are:

a. Kenmore Automatic Washing Machine Model 600, or similar washing machine. In case of dispute, the Kenmore Model in use at the procurement activity shall be used.

b. Kenmore Automatic Dryer Model 600, or equivalent.

c. "Formatic" hot head pressing machine or equal. "Formatic" manufactured by the American Laundery Machine Co., Cincinnati, OH.

6.8 Fusing conditions. The following fusing conditions have been found to satisfactorily obtain the required results of this specification. These limits are furnished for guidance only. For optimum results with the particular fusible used, specific machine settings should be obtained from the contractor furnishing the fusible material.

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	Minimum	Maximum
Temperature (between pressing surfaces), °F	320	370
Pressure (between pressing surfaces), PSI	20	40
Pressing time, seconds	14	20

* 6.9 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activity:

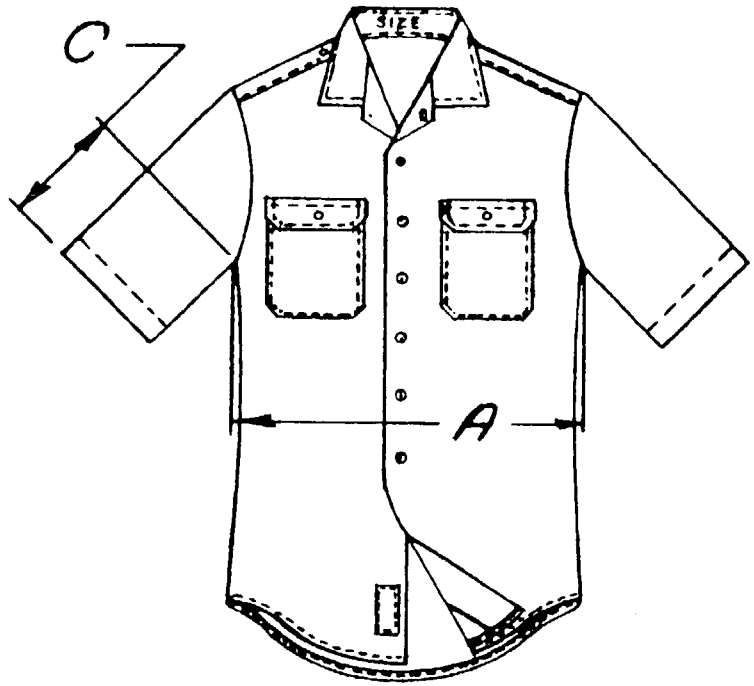
Army - MD
DLA - CT

Preparing activity:

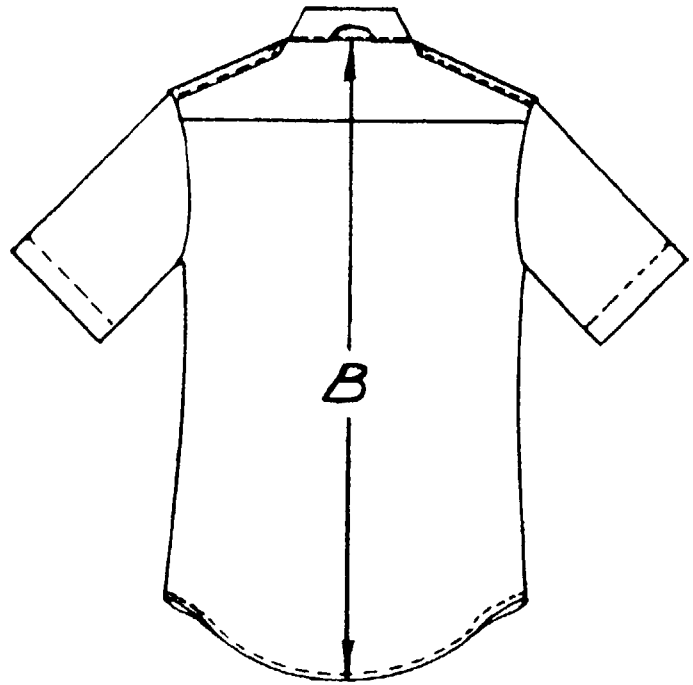
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FRONT



BACK

FIG-1 SHIRT, MAN'S, SHORT SLEEVE, ARMY GREEN 415, DURABLE PRESS

2-1-1497

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-44041A(GL) Shirt, Man's, Short Sleeve, Polyester/Cotton, Army Green 415, Durable Press

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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